

Advanced (Habanero)

- Q1.** Solve the system: $x^2 + y^2 = 4$, $x^2 - y^2 = -1$
(A) $(-1, -\sqrt{3})$
(B) $(-1, \sqrt{3})$
(C) $(1, -\sqrt{3})$
(D) $(1, \sqrt{3})$
(E) No solution
- Q2.** What is the graphical representation of $x^2 + y^2 = 4$?
(A) Line
(B) Circle
(C) Parabola
(D) Ellipses
(E) Hyperbola
- Q3.** Solve: $y = x^2 - 4$, $y = -x^2 + 4$
(A) $(0, 4)$
(B) $(-2, 0)$
(C) $(2, 0)$
(D) $(-2, 0)$ and $(2, 0)$
(E) No solution
- Q4.** Find the intersection points of $x^2 - y^2 = 0$, $x^2 + y^2 = 4$
(A) $(0, 0)$
(B) $(-2, 0)$
(C) $(2, 0)$
(D) $(-2, 0)$ and $(2, 0)$
(E) No solution
- Q5.** The system $x^2 + y^2 = 1$, $x - y = 1$ has
(A) One solution
(B) Two solutions
(C) No solution
(D) Infinitely many solutions
(E) Cannot be determined
- Q6.** Graphically, what does the solution to a system of quadratic equations represent?
(A) Intersection points
(B) Parallel lines
(C) Perpendicular lines
(D) Asymptotes
(E) Axis of symmetry
- Q7.** What method can be used to solve $x^2 + y^2 = 4$, $y = x^2 - 4$?
(A) Substitution
(B) Elimination
(C) Graphical
(D) Both A and C
(E) All of the above
- Q8.** The system $y = x^2 - 2$, $y = -x^2 + 2$ has solutions
(A) At $(0, 2)$ and $(0, -2)$
(B) At $(-1, 1)$ and $(1, 1)$
(C) At $(-\sqrt{2}, 0)$ and $(\sqrt{2}, 0)$
(D) At $(0, 0)$
(E) No solution

Open Ended Questions (FRQ)

- Q9.** Solve the system $x^2 - y^2 = 3$, $x^2 + y^2 = 8$
- Q10.** Find the points of intersection between $y = x^2 - 1$ and $y = -x^2 + 4$. Show your work.

Chef's Tips

- Emphasize the importance of checking solutions in the original equations.
- Note that graphical solutions may involve using a graphing calculator or other technology.
- Remind students to consider all possible methods of solution, including algebraic and graphical approaches.